

Class 5: Integrated Sensing and Communication

What it is, why it is hard, and how OFDM does it

Xin Liu

Florida State University

xliu15@fsu.edu

CIS 4930/5930 Future Edge Networks

xinliulab.github.io/FSU-CIS4930-CIS5930-Future-Edge-Networks

Two machines, one sky



Frankfurt airport surveillance radar

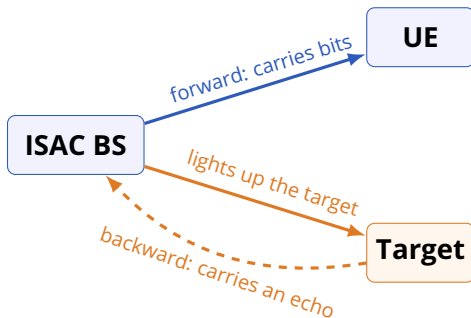


Ericsson active 5G antennas

Q: One sends a wave to be understood. One sends a wave to come back. What do they share?

A: Spectrum, power, antennas, and silicon — all of it scarce. Building two systems that never talk to each other pays for the same hardware twice.

One waveform, two jobs



The same transmission is **information** on the way out and **measurement** on the way back.

ISAC does not add a radar to the network.
It reads the network's own signal as radar.

An echo is data nobody uploaded



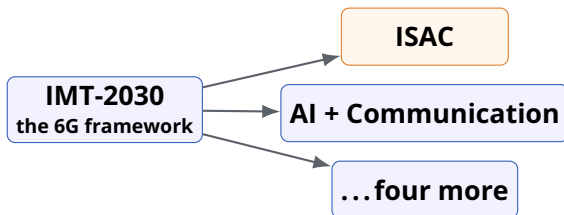
Waymo Jaguar I-Pace roof sensor package

- **Vehicles:** see what the camera cannot.
- **XR:** track a hand, no wearable.
- **IoT:** count people, no camera.
- **Space:** range the satellite you talk to.
- **Safety:** a drone, a fall, a flood.
- **Farming, weather, health:** same echo.

Every one of these is the network noticing something on its own.

XR = Extended Reality IoT = Internet of Things

Sensing is written into the 6G frame



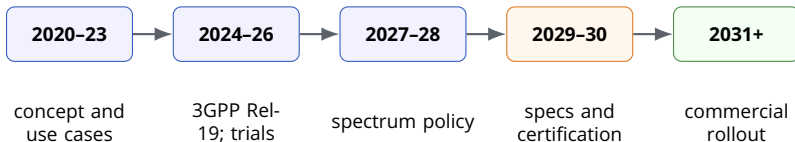
Q: Why does a standards body have to say this out loud?

A: Because a usage scenario is what gets spectrum, performance requirements, and test procedures written for it. Until ISAC was named, no radio was obliged to be measurable.

ITU-R Recommendation M.2160, IMT-2030 framework (2023)

Being named starts a clock

Being written into M.2160 does not ship a product; it starts a schedule — spectrum, requirements, tests, then certification. Here is where that schedule stood when this material was prepared.

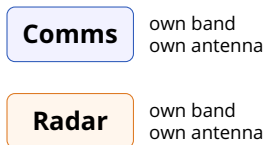


The 2020–2026 columns describe work that has happened. Everything from 2027 rightward is a forecast made in this talk, not a published schedule.

Standardisation is where the tradeoff stops being a curve and starts being a number somebody must certify.

Two boxes became one box

Separate systems



DFRC



What it buys

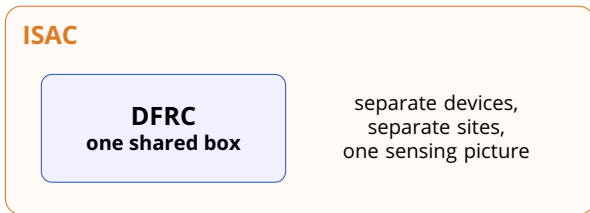
One spectrum licence, one amplifier, one aperture, one power budget.

DFRC = Dual-Function Radar Communication

What it costs

A single waveform now has to satisfy two specifications at once.

DFRC is one box. ISAC is the question.

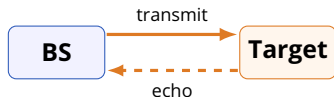


DFRC shares waveforms,
antennas, and processing chains
inside one radio.

ISAC also lets one node transmit
and a different node listen.

Every DFRC system is ISAC.
Not every ISAC system fits in one box.

What an echo actually tells you



- **Delay** → range, $R = c\tau/2$
- **Doppler** → closing speed
- **Angle** → direction

Q: Which knob buys finer range, and which buys finer speed?

A: Bandwidth buys range resolution — a wider signal has a sharper peak in time. **Observation time** buys velocity resolution — you must watch long enough for the phase to drift.

Three questions for the rest of today

1 Why can't one signal be great at both jobs?

The two jobs want opposite things. That gives us a curve of compromises, not a fix.

2 Which signal do we actually use?

OFDM — the one Wi-Fi, 4G and 5G already transmit. No new radio.

3 How do we share it between bits and echoes?

The OFDM grid is a budget. Somebody has to decide who gets each cell.

Everything from here answers one of these three.
If you lose the thread, ask which one we are on.

They want opposite things (1/2)

	Sensing wants	Communication wants
Spectrum	Wide bandwidth, for resolution	Efficient bandwidth, for rate
Signal	Deterministic waveforms	Random, information-bearing symbols
Power	High transmit power on the target	Power efficiency per bit
Time	Continuous illumination	Low-latency bursts
Space	Wide or scanning beams	Narrow beams, one per user

Randomness is what carries information — and randomness ruins a clean echo.
That single sentence is the whole conflict.

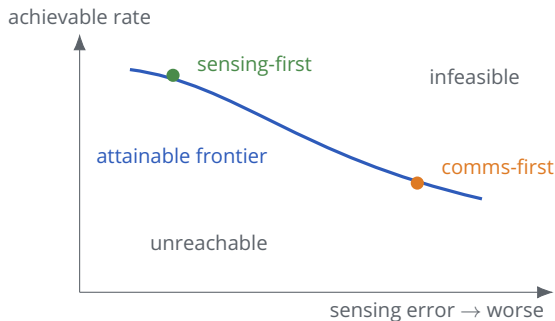
They want opposite things (2/2)

Sensing wants		Communication wants
Interference	A clean echo, no clutter	A clean constellation, no radar leakage
Domain	Fine steps in distance and speed	Fine slots in time and frequency
Hardware	Precise timing, high-power amplifiers	Spectral efficiency, many users
Latency	Low-latency tracking feedback	Tolerates delay off the critical path
Regulation	Frequency allocation rules	Emission and efficiency standards

Q: Is there a setting where both are optimal?

A: No. There is a **Pareto frontier**: past a point, every improvement in one is paid for out of the other. Design work is choosing where to sit on it, not escaping it.

The frontier, drawn once



Horizontal axis = sensing error: how far off the range and speed estimates are. **Lower is better**, so the good corner is up and to the left.

Every result in the rest of this lecture is a claim about one point on a curve like this.

Question 2: which signal do we actually use?

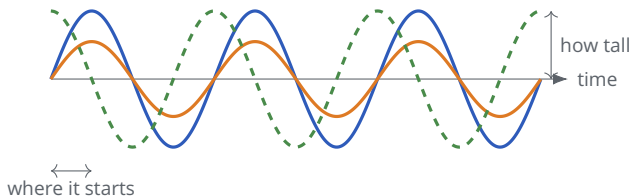
Question 1 is answered: the two jobs want opposite things, so there is a curve of compromises. Whatever signal we pick has to carry **both** jobs and sit somewhere on that curve.

The answer is OFDM (Orthogonal Frequency Division Multiplexing) — the signal Wi-Fi, 4G and 5G already transmit. No new radio.

Before we can read OFDM as a radar we need to understand it properly. Most of you have only heard the name, so we build it from the ground up, in four parts:

- 1 What a radio actually sends — and why that is a point.
- 2 Why we send many waves at once, and why they must be perpendicular.
- 3 Why the FFT — the machine that adds waves up and takes them apart.
- 4 Putting it together: transmitter, receiver, and the grid.

A radio sends a wave, not bits

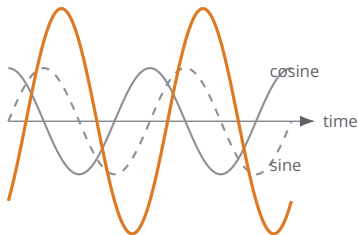


blue: tall, starts at 0 **orange:** half as tall **green:** tall, starts a quarter cycle early

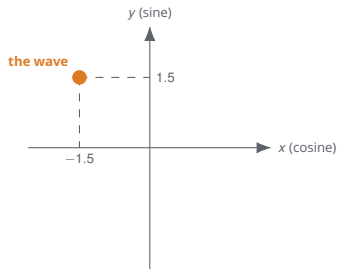
An antenna cannot emit a 0 or a 1. It emits a **wave** at some frequency. At one frequency a wave has exactly two things you can change: **how tall it is** (amplitude) and **where in its cycle it starts** (phase). Everything else about it is fixed by the frequency.

Two knobs. To send information, we turn the knobs.

Two knobs are one point



$$\text{orange} = -1.5 \times \text{cosine} + 1.5 \times \text{sine}$$



Any wave of the chosen frequency — any height, any starting point — can be written as “x parts of a cosine plus y parts of a sine”. So the two knobs are just two numbers, and two numbers are a **point** (x, y).

One point = one wave. From now on, when you see a dot, think “a wave”.

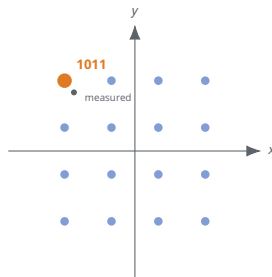
Bits pick the point

Agree on 16 allowed points — 16 allowed waves (the scheme is called 16-QAM; the name does not matter). Cut the bits into groups of four:

1011 0010 1110 0001 ...

Each group names one point. The radio sends that wave for one **symbol time** T , then the next group's wave, and so on.

The receiver measures (x, y) , picks the **nearest** allowed point, and reads four bits back. Noise nudges the measurement, so allowed points must keep their distance — hence 16, not thousands.



16 allowed waves, 4 bits each

One wave carries 4 bits every T : slow.

Part 2: one wave is slow — so use many

Four bits per symbol time is nowhere near a video stream. Two ways to go faster:

Option A: one wave, shorter symbols.

Keep one wave and switch points 64 times faster. Symbol time shrinks from T to $T/64$.

Option B: 64 waves at once.

Use 64 different frequencies at the same time, each carrying its own point for the full time T . Add them up and send the sum.

Both give 64 times the bits. Every modern radio — Wi-Fi, 4G, 5G — chose **B**. The next slide is the reason, and it is the same physics this whole lecture is about.

Same speed. So why does everyone pick many slow waves over one fast one?

Why not one fast wave? Echoes.

A: short symbols



echo lands on the next symbol

B: long symbols



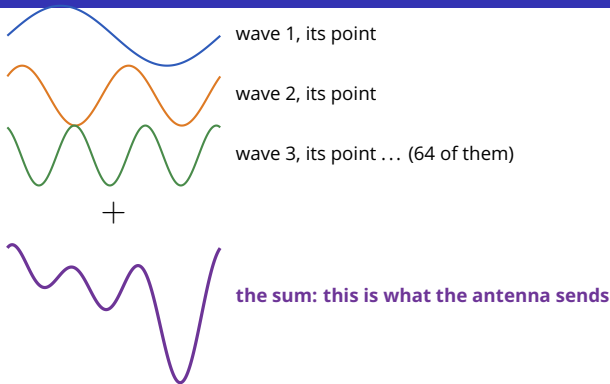
echo mostly lands on itself

↔
echo delay

Radio bounces. A copy of your signal arrives a little late off a wall or a building — about $1\ \mu\text{s}$ later for every 300 m of extra path. With short symbols (A) that echo is a smeared copy of the previous symbol on top of the current one: garbage. With long symbols (B) the echo overlaps the same symbol almost entirely; it just makes the wave a bit taller or shorter, which the receiver can undo.

Echoes are why we send many slow waves instead of one fast one.

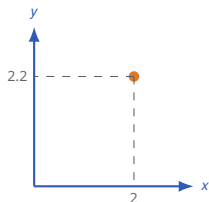
Add 64 waves and send the sum — then what?



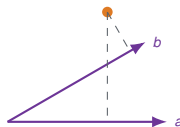
The receiver gets only the purple line. It has to recover **each** wave's (x, y) from it — 64 pairs of numbers out of one squiggle. Looking at the squiggle, that seems hopeless.

When can you pull one wave back out of a sum of many?
That is the question the word "orthogonal" answers.

Perpendicular axes: reading one ignores the other



perpendicular: reading x
picks up nothing of y



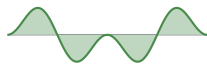
tilted: reading a
picks up part of b

Think of a point on a plane. With perpendicular axes, its x -reading does not change when you slide it along y : the two numbers are independent, each can be read on its own. Tilt the axes and that breaks — moving along b changes the a -reading. The two numbers leak into each other.

Separable $\Leftrightarrow$ perpendicular. Now we need the same idea for waves.

Perpendicular waves, drawn

1 cycle $\times$ 2 cycles in time T



product

area above = area below: total 0

1 cycle $\times$ 1.5 cycles in time T



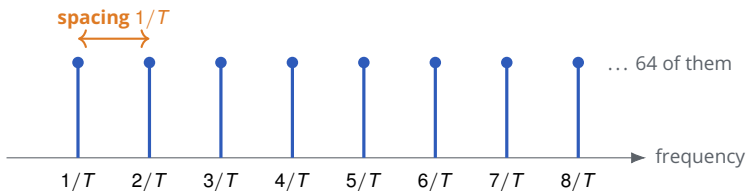
product

more above than below: total $\neq 0$

To measure “how much of wave 1 is in the sum”, the receiver multiplies the sum by wave 1 and adds up over the symbol time T . Two waves are **perpendicular** when that product adds up to **zero**: measuring one picks up nothing of the other. Left: it works. Right: it does not — wave 2 leaks into the reading.

Whole cycles in T : perpendicular. Fractional cycles: leak.

The perpendicular family: frequencies k/T

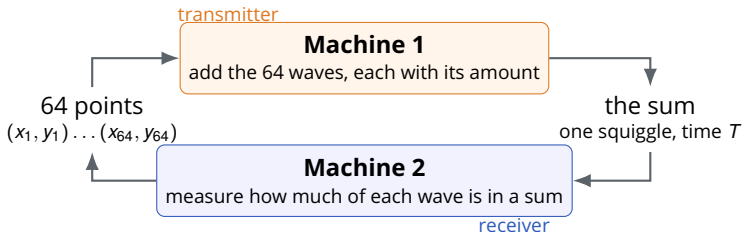


A wave fits a whole number of cycles into T exactly when its frequency is a multiple of $1/T$. So the waves at $1/T$, $2/T$, $3/T$, ... are **all perpendicular to each other**. That is the family OFDM uses. Each member is called a **subcarrier**, and their spacing is $1/T$ — not a design choice, a consequence.

Real numbers. 5G: $T \approx 67 \mu\text{s}$, so subcarriers 15 kHz apart (30, 60, 120 kHz variants exist). Wi-Fi: $T = 3.2 \mu\text{s}$, spacing 312.5 kHz.

We now know which 64 waves to use.

Part 3: we need two machines



Both machines work on the same 64 waves. To put them in a computer, sample each wave at 64 instants during T : a wave becomes a row of 64 numbers, and 64 waves become a 64×64 table. **Machine 2 is that table.** Machine 1 is the same table run backwards.

That table has a name: the FFT matrix.

Two waves: the 2×2 FFT, with numbers

Machine 2 — measure. Two waves sampled at two instants: a flat one (1, 1) and a flipping one (1, -1). Stack them as rows:

$$\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \end{pmatrix}$$

The sum (3, 1) contains 4 units of the flat wave and 2 of the flipping one. Rows perpendicular:
 $1 \cdot 1 + 1 \cdot (-1) = 0$.

Machine 1 — add. Same table, run backwards (divide by 2):

$$\frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 4 \\ 2 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$$

Give it the amounts (4, 2) and it rebuilds the sum (3, 1). Round trip complete, nothing lost — because the rows are perpendicular.

FFT = measure each wave. IFFT = add the waves back.
Perpendicular rows are what make the round trip exact.

Four waves: the 4×4 FFT

Four waves at $0, 1/T, 2/T, 3/T$, each sampled at four instants:

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{pmatrix}$$

Row k is the wave at frequency k/T . The $j (= \sqrt{-1})$ is only there so a wave can rotate through its cycle instead of merely flipping sign.

Check one pair. Row 2 against row 4 (flip the sign of j in one of them, multiply entry by entry, add):

$$\begin{aligned} 1 \cdot 1 + (-j)(-j) + (-1)(-1) + j \cdot j \\ = 1 - 1 + 1 - 1 = 0 \end{aligned}$$

Zero. Try any other pair — also zero. **Every wave is perpendicular to every other**, exactly as the drawing promised.

The 64×64 table is the same thing, bigger. The 4096×4096 one too.

The FFT matrix is the list of perpendicular waves, written as numbers.

Why “Fast” is the important letter

Multiplying a signal by an $N \times N$ table costs N^2 multiplications. The FFT gets the identical answer in about $N \log_2 N$ — by reusing the halves of the table, which are copies of each other.

waves N	plain table N^2	FFT $N \log_2 N$	saving
64	4 096	384	11×
1 024	1 048 576	10 240	100×
4 096	16 777 216	49 152	340×

A 5G phone runs this for every symbol, tens of thousands of times a second, on a battery. With the plain table it could not. The FFT algorithm dates from 1965; the first paper proposing OFDM built on it is from 1971. That is not a coincidence.

OFDM is practical because the FFT is fast.

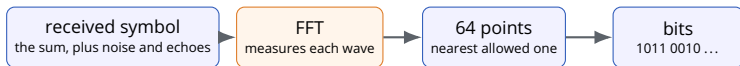
Part 4: the transmitter



- Take 256 bits. Cut into 64 groups of four; each group picks a point.
- Put point k on the wave at frequency k/T — on **subcarrier** k .
- One IFFT adds the 64 waves into one burst of length T : the **symbol**.
- Send it. Then the next 256 bits, the next symbol, and so on.

256 bits in, one symbol out — 64 times what one wave could do.

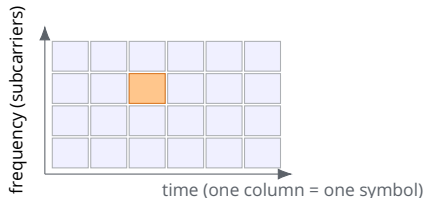
The receiver



- One FFT on the received burst: out come 64 (x, y) pairs, one per subcarrier — cleanly, because the waves are perpendicular.
- Each pair is nudged by noise; snap it to the nearest allowed point; read 4 bits.
- The echo from slide “Why not one fast wave” is in here too. It makes each subcarrier’s point a little larger or rotated — a per-subcarrier correction, then forgotten. Or kept: that echo is what the radar half of ISAC reads.

Same 64 waves, same FFT, read the other way round.

Symbol after symbol: the grid

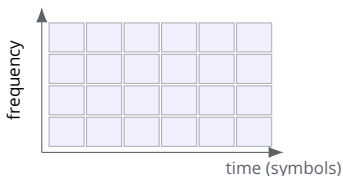


- Each column is one symbol; each row is one subcarrier.
- The orange cell: subcarrier 3 during symbol 3 — **one wave, one symbol time, one point, four bits.**
- 5G calls a cell a **resource element**. A base station schedules by handing out cells: these to you, those to your neighbour.

Every OFDM radio — Wi-Fi, 4G, 5G — fills and reads this grid, symbol after symbol.

Question 2 answered: OFDM is a grid of cells, one wave per cell.
Next slide: the same grid, read as a radar.

OFDM already carries a radar inside it



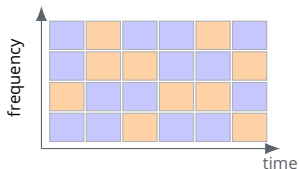
- Mature: Wi-Fi, 4G, 5G, 6G.
- The grid is already knobs.
- A cell serves bits or echoes.
- No new modulator needed.

Q: If OFDM is already good enough, what is left to design?

A: Which cells go to bits and which go to echoes. Every cell handed to sensing is a cell taken from data.

OFDM = Orthogonal Frequency Division Multiplexing

The grid is the budget



■ communication RE ■ sensing RE

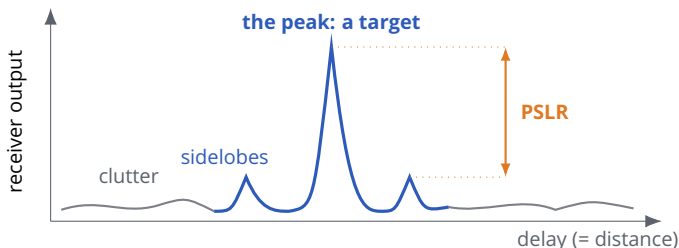
- One RE is one cell.
- Each carries symbol and power.
- Assign cell, then power.
- Two functions, one grid.

RE = Resource Element, one subcarrier during one OFDM symbol

PSLR = Peak-to-SideLobe Ratio; RoI = Region of Interest: the distances and speeds we actually care about

Cross-Domain Dual-Functional OFDM Waveform Design, IEEE JSAC 2024

What a peak is, before we grade one



The receiver slides a copy of its own waveform across the echo. Where a target sits, everything lines up and adds — that spike is **the peak**. **Sidelobes** are ghosts of our own waveform; **clutter** is every echo we never asked for.

PSLR = how far the true peak stands above the tallest ghost.
That number is the sensing grade on the next slide.

Two scores, graded in two rooms

Communication score

Achievable data rate, summed over every RE assigned to communication.

Measured in **time-frequency**: put bits where the channel is good.

Sensing score

PSLR inside the region of interest — how far the true peak stands above the clutter.

Measured in **distance and speed**: make the peak stand tall where the targets actually are.

The same grid is graded twice,
by two examiners in two different domains.

Whoever picks first, wins

Communication-centric

- 1 Give the good REs to bits.
- 2 Then tune leftover powers.
- 3 Sensing takes the remainder.

Rate stays optimal; PS LR is merely good inside the RoI.

Sensing-centric

- 1 Lock the peak shape inside the RoI.
- 2 Then tune the rest of the grid.
- 3 Bits take the remainder.

PS LR is locally perfect; rate approaches, not reaches, optimal.

Q: Both designs claim “almost no loss”. How can that be?

A: Because the constraint is only enforced inside the **region of interest**. Narrow the room you must be perfect in, and the rest of the grid is free for the other function.

Three things to take home

- 1 **One waveform, two jobs.** The signal a base station already sends comes back as an echo, and that echo is measurement nobody had to upload.
- 2 **The two jobs want opposite things.** Sensing wants a wide, steady, predictable signal; data wants an efficient, bursty, random one. So there is a curve of compromises, not a fix.
- 3 **On OFDM, picking a point on that curve is a scheduling decision.** Which cells of the grid carry bits, which carry echoes, and who picks first.

ISAC is not a new radio.

It is a new question asked of the radio we already have: who gets each cell?

What is actually unfinished

Settled enough to build

- The tradeoff is understood.
- OFDM-based ISAC works.
- The RE grid can be allocated.

Still open

- A killer application.
- Security of a shared waveform.
- AI for waveform design.
- Space-air-ground reach.

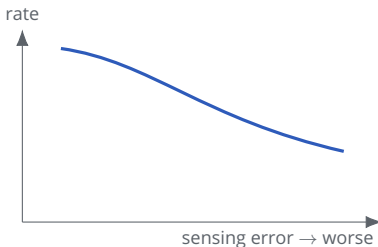
Q: Which of these is not an engineering problem?

A: The first. A waveform can be optimised; a market cannot. ISAC has a technology looking for the application that pays for it.

SAGIN = Space-Air-Ground Integrated Network

Before you go: pick a point on the curve

Two minutes, on paper, handed in on the way out. Same curve as before.



- 1 Mark where a **self-driving car** should sit. Mark where a **home Wi-Fi link** should sit.
- 2 Name one resource the two designs would allocate differently.
- 3 Answer in one sentence:
what does sensing cost the bits in your example?

Appendix: the acronym decoder ring

Area	Term	Meaning / why it matters
System	ISAC	Integrated Sensing and Communication
System	DFRC	Dual-Function Radar Communication; ISAC inside one radio
System	IMT-2030	The ITU framework naming ISAC as a 6G usage scenario
Waveform	OFDM	Orthogonal Frequency Division Multiplexing
Metric	PSLR	Peak-to-SideLobe Ratio; true peak above the clutter
Metric	SINR	Signal to Interference-plus-Noise Ratio
Resource	RE / RoI	Resource Element (one grid cell); Region of Interest (the distances and speeds we care about)

Appendix: the quantities behind every plot

Delay τ range, $R = c\tau/2$

Doppler ν speed, $v = \nu c/(2f_c)$

Bandwidth buys range resolution

Dwell time buys velocity resolution

Rate $\sum \log_2(1 + \text{SINR})$ over REs

Sensing error has a hard floor set by bandwidth, power and time

PSLR peak over worst sidelobe, in the RoI

Rate is a sum you maximise.
Sensing error has a floor you cannot go below.

Sensing accuracy is bounded from below.
That is why the frontier bends.

- **Zhu Han (Univ. of Houston), “Integrated Sensing and Communication for 6G”**

The invited talk this lecture is built from. Slides and code: wireless.egr.uh.edu.

- **ITU-R M.2160: IMT-2030 framework**

Where ISAC becomes an official 6G usage scenario.

- **Cross-domain dual-functional OFDM waveform design**

IEEE JSAC 2024. Source of the RE-allocation results.